



# Online Sexual Interventions for Enriching Couples' Sex Lives: A Narrative Review

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## Abstract

**Purpose of Review** The objectives of this narrative review are to describe existing research and to examine the effectiveness of online interventions used to address sexual dysfunction and improve couples' sex lives.

**Recent Findings** Following the PRISMA guidelines, we conducted a narrative review to synthesize the evidence for the efficacy of online sexual intervention from experimental studies in the past five years. After removing duplicates, 554 articles were screened for eligibility, and 17 fulfilled the inclusion criteria for the review. Ten studies were conducted in North America or Europe, with the rest focusing on Asia. Participants were mostly young and middle-aged adults, with the average age ranging from 25 to 46 years, except for one study in which the median participant age was 60. Some study samples were drawn from clinical populations with a diagnosed condition (e.g., sexual dysfunction, infertility, or related concerns), while others were recruited from the general population. Eleven studies utilized online videos, including psychoeducation about sex, sexual communication, and problem-solving skills. Six studies had therapists or coaches provide feedback or answer questions in between online sessions. The effectiveness of the online intervention varied significantly across studies, with the unified effect sizes ranging from 0.02 to 1.76. For the outcome measures specifically assessing sexual function, 2 out of the 23 outcomes had large effects, 9 had moderate effects, and 12 had small effects. On the other hand, measures of relationship satisfaction and communication showed greater improvements, with 21 out of the 32 outcomes (66%) having moderate effects and 2 outcomes having a large effect.

**Summary** The effectiveness of online sexual intervention varied across the studied population, measured outcomes, and the intervention techniques, likely reflecting heterogeneity in baseline sexual function and intervention intensity. Interventions were more effective in improving sexual communication and relationship satisfaction than directly addressing sexual dysfunction. Future research should clarify underlying mechanisms to optimize scalable, tailored online interventions for enriching couples' sex lives.

**Keywords** Online intervention · Sexual function · Relationship · Sexual satisfaction

## Introduction

Sexual satisfaction is fundamental to couple dynamics, and has strong, positive associations with relationship quality, satisfaction, and stability across different populations [1–6].

Even for couples experiencing life stressors, sexual satisfaction also appears to be an important predictor of relationship satisfaction. For example, among women with ovarian cancer, better sexual satisfaction was associated with increased marital satisfaction for both themselves and their partner [7].

One of the key contributors to sexual satisfaction is sexual functioning, which encompasses physiological responses such as desire and arousal, as well as the ability to experience sexual pleasure. However, the prevalence of sexual dysfunction worldwide is high, affecting about 15% to 43% of men and 23% to 46% of women, depending on age group and diagnostic criteria [8–12]. This underscores the need for effective treatments for sexual dysfunction,

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since it not only leads to dissatisfaction and distress in relationships, but also has negative impacts on an individual's well-being, such as increased rates of depression and anxiety, reduced sexual self-esteem, and feelings of guilt and frustration [13–15]. Indeed, low levels of sexual satisfaction are a common focus in couples therapy [16, 17]. Taken together, existing research suggests there is an emerging need for widely accessible interventions to treat sexual dysfunction and improve couples' sex lives.

Within the mental health field, there has been a relatively recent trend to integrate technology into treatment. For example, England's National Health Service (NHS) started to test computerized mental health technologies to treat anxiety and depression over the last two decades [18]. When it comes to issues related to sex, online interventions may be particularly beneficial because they mitigate self-consciousness, shame, and discomfort associated with sex that may hinder help-seeking and treatment effectiveness [14, 18]. Anonymity encourages treatment initiation by those in need of sex therapy, but who otherwise might avoid traditional in-person face-to-face treatment. Online self-help resources are also available at any time and place, can be utilized by many individuals in different locations at the same time, and can be reviewed as often as people want, making the format ideal for interventions to be implemented widely with a lower cost [19, 20]. While there has been an increase in promoting digital interventions to address sexual problems, existing systematic reviews have largely examined online interventions targeting sexual dysfunction at the individual level, with couple involvement, when present, treated as a secondary or optional component (Srisadono et al.'s [21] review only focused on male sexual dysfunction and Ferreira et al.'s [22] review was broadly about treating sexual dysfunction for individuals). Other reviews have evaluated digital interventions for couples' relationship satisfaction broadly, without a specific focus on sexual outcomes [23]. To date, no review has synthesized evidence on online interventions delivered at the dyadic level that jointly target sexual functioning and relationship satisfaction. As such, the current review aimed to describe recent experimental research (within the past five years) and examine the effectiveness of online interventions in addressing sexual dysfunction and improving couples' sex lives, with particular attention to how intervention characteristics and population factors relate to outcomes across both relationship and domains.

## Methods

The present narrative review was conducted following the guidelines of the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) statement [24].

Searches were conducted in the following three web-based databases: PubMed, PsycINFO, and Web of Science. The initial search was conducted on August 12, 2026, and studies published in the preceding five years were included in the review. Further inclusion criteria were (1) an experimental study (including randomized controlled trials, clinical trials, pilot studies, or feasibility studies), (2) using an online psychological intervention, (3) to address sexual problems or improve sex lives, and (4) in which couples completed at least some components of the intervention together. The search terms combining the four domains were presented in Table 1 in Supplementary Material. Articles that were not in English or not peer-reviewed were excluded.

Titles and abstracts were first screened for preliminary eligibility using the web-based platform Covidence [25]. Full-text screening was then completed to ensure that all included studies met the eligibility criteria.

Data were extracted and synthesized based on three domains for each study: (1) study characteristics, including title, authors, publication year, country, sample size and demographics, and inclusion and exclusion criteria; (2) intervention characteristics, including intervention components, duration, and delivery; and (3) study results, including outcomes related to sexual (dys)function as well as sexual and relationship satisfaction.

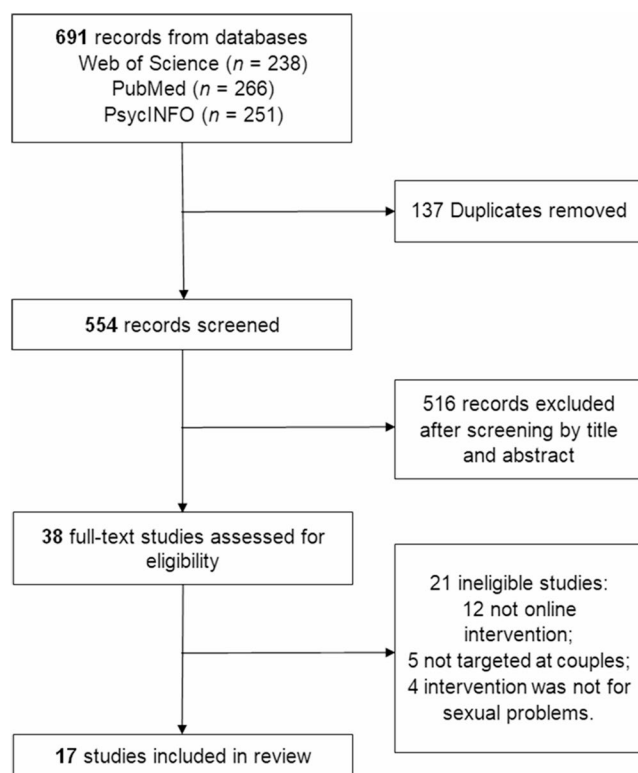
## Results

We retrieved in total 691 study records from the databases. After removing duplicates, a total of 554 unique titles and abstracts were available for the initial screening. Following the initial screening, the full texts of 38 studies were assessed for eligibility. Finally, 17 studies met the eligibility criteria and were included in the review. The 21 ineligible studies were excluded for one of three reasons: (1) the intervention was delivered face-to-face/onsite rather than online ( $n=12$ ); (2) the intervention recruited individuals rather than couples ( $n=5$ ); or (3) the intervention was not designed to address sexual problems (e.g., targeting couples' financial stress, condom use, or general communication) ( $n=4$ ).

Figure 1 presents the PRISMA flow diagram of the search and selection process.

## Study Characteristics

Table 2 in Supplementary Material shows the characteristics of the studies we included. Ten studies were conducted in North America or Europe, while the rest were conducted in China, Iran, South Korea, and Turkey. Participants were all adults, with the average age ranging from 25 to 46 years, except for one study in which the median participant age



**Fig. 1** The PRISMA Flow Diagram Outlining the Search and Selection Process

was 60. The specific foci of the studies varied depending on the study aims and the target population. For instance, some aimed at addressing sexual problems and/or relationship distress, while others focused on better coping due to life stressors. While five studies included same-sex/gender diverse couples in the intervention and the study by Nowlan and colleagues [26] only looked at same-sex couples, overall, most of the participants were in a different-sex/gender relationship.

Fourteen out of the 17 studies employed a randomized controlled trial study design. Two studies by Bouchard and colleagues [27] and Nowlan and colleagues [26] were a feasibility study and a pilot study without a control group, and Kleinplatz and colleagues [28] used a quasi-experimental approach without randomization when assigning the experimental and waitlist control groups.

### Intervention Components and Delivery

Table 1 presents the intervention characteristics of the reviewed studies, including the components, delivery format, and the duration of the intervention. Studies by Hatch and colleagues [29], Le and colleagues [30], and Roddy and colleagues [31] were discussed together because they all utilized the OurRelationship and ePREP intervention programs. Similarly, studies by Huang and colleagues [32] and

Li and colleagues [33] were also examined together, as both employed an online animated Sensate Focus intervention.

The intervention duration ranged from 4 days to 28 weeks. Eleven studies used online videos in some form, covering materials like psychoeducation about sexual communication and sexual health, facilitating the understanding of each other, and improving their problem-solving skills. Among them, four studies [32, 42] incorporated sensate focus techniques as part of intervention; six studies [26, 29, 31, 34, 40] had therapists or coaches intervene in-between online automated sessions, such as providing feedback of the assigned practice and addressing questions. Additionally, the study by Durbin and colleagues [35] employed a smart-phone application to prompt participants to demonstrate physical affection to their partner as the intervention, without any formal psychosexual-based components.

While the remaining five studies [27, 28, 36, 41] utilized online platforms to provide the intervention, they all conducted couples therapy or group sexual counseling as the intervention, instead of online, video-based, self-learning modules.

### Intervention Outcomes

Table 2 shows the outcome concept and the measurements that each study used, as well as the statistical results reported in the original article. To quantitatively synthesize the results across studies, all outcomes were converted to a standardized mean differences (SMD) metric, expressed as Hedges' *g*. Hedges' *g* corrects Cohen's *d* for small-sample bias, and positive values of *g* indicate greater improvement in the intervention group. For each study, the most appropriate statistic was extracted (e.g., *time*  $\times$  *group* interaction or between-group post-test comparison for RCTs, within-group pre–post comparison for feasibility studies). Where possible, means and standard deviations (SDs) at baseline and post-test were used; otherwise, reported *t*, *F*, regression coefficients (*b*), odds ratios (ORs), or raw Cohen's *d* values were converted.

Across the fourteen reviewed RCTs and the one quasi-experimental study, where the intervention effect was measured by *time*  $\times$  *group* interaction or between-group post-test comparison, Hedges' *g* varied considerably, ranging from 0.02 to 1.76 depending on different studies with different outcomes. For the feasibility study and the pilot study, where the intervention effect was measured by within-group pre–post comparison, Hedges' *g* varied from 0.3 to 1.05, indicating moderate to high effects.

**Outcome Measures** Among all the assessed outcomes regarding sexual functioning, the median effect was 0.18, ranging from 0.02 to 1.75, with 52% of the measures

**Table 1** Intervention characteristics

| Reference                   | Components                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Delivery formats                                                                                                                                   | Duration                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Bouchard et al. [27]        | Cognitive-behavioral couple therapy (CBCT) intervention: manual-based, 12-session adapted for genito-pelvic pain/penetration disorder, also focused on the interpersonal factors such as sexual motivation, sexual communication, relational self-expansion, and emotion co-regulation.                                                                                                                                                                                                                                                                                           | Online couple therapy led by four therapists.                                                                                                      | 18 weeks                                               |
| Dreischor et al. [34]       | The Pleasure & Pregnancy (P&P) program: 8 modules about psychosexual information and couple communication, mindfulness and sensate focus exercises aiming to improve sexual pleasure and hence increase pregnancy rates.                                                                                                                                                                                                                                                                                                                                                          | Online program with mixed media (texts, videos, audio recordings, and drawings), online interaction with peers and via email with coaches          | 24 weeks                                               |
| Durbin et al. [35]          | A smartphone application: randomly prompted participants to demonstrate physical affection to their partner.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Smart-phone application prompts                                                                                                                    | 2 weeks                                                |
| Farahi et al. [36]          | Group A: women by themselves; Group B: women participating with their husband in sessions 2 and 3. The intervention in both groups were weekly group sexual counseling sessions lasting 2 h each, focusing on the of the “Good Enough Sex” (GES) model, plus exercises and individual feedback at the end of each session.                                                                                                                                                                                                                                                        | Online group sexual counseling sessions, weekly contact with participants via phone calls to address questions.                                    | 4 weeks                                                |
| Hatch et al. [29]           | Group 1: OurRelationship program, consisting of 3 sections: “Observe,” where individuals identify relationship problems to focus; “Understand,” where individuals achieve a more accurate understanding of the problem; “Respond,” where partners developed a plan for ameliorating the problem. All content lasted approximately 7 h. Group 2: the computer-based relationship focused preventive intervention (ePREP) [37]: teaches couples how to recognize and combat risk factors that lead to relationship distress, specifically communication and problem-solving skills. | Group 1: online program of 8 h and interspersed with four 15 min videoconference calls with a staff coach. Group 2: online program and coach calls | 8 weeks                                                |
| Heidari et al. [38]         | A structured couple counseling program consisted of five sessions, including psychosexual education about anatomy, sexual response, postpartum changes; skills-building such as proper nutrition and mindfulness practices, addressing sexual dysfunctions, introducing Kegel exercises; and cultivating communication and emotion regulation skills.                                                                                                                                                                                                                             | Online synchronous counseling sessions led by a counselor via voice or video calls.                                                                | Unspecified, each session lasted between 30 to 45 min. |
| Huang et al. [32]           | 11 animation videos based on the Sensate Focus sex therapy, including self-focus exercises such as the Kegel exercise, as well as couple-level exercises such as gazing, hugging, and non-genital touching.                                                                                                                                                                                                                                                                                                                                                                       | Online animation videos                                                                                                                            | 5 weeks                                                |
| İnan Kırnızgöl & Şahin [39] | Couples completed two modules either as a couple or individually: Module 1: Communication and Module 2: Sexual and Reproductive Health, consisting of 9 video segments totaling 194 min.                                                                                                                                                                                                                                                                                                                                                                                          | Online videos                                                                                                                                      | Unspecified, approximately 4 to 12 weeks               |
| Kleinplatz et al. [28]      | “Optimizing Sexual Experiences” (OSE) group couple therapy: psychoeducation about sexual pleasure and experimental exercises, tailored to meet couples’ needs.                                                                                                                                                                                                                                                                                                                                                                                                                    | Group couples sex therapy (online during COVID, in-person prior to COVID)                                                                          | 4 weeks (2 h per week)                                 |
| Le et al. [30]              | Same as Hatch et al. [29]: two groups going through OurRelationship program and ePREP.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Group 1: online program of 8 h and interspersed with four 15 min videoconference calls with a staff coach. Group 2: online program and coach calls | 8 weeks                                                |
| Li et al. [33]              | Same as Huang et al. [32]: 11 animation videos based on the Sensate Focus sex therapy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Online animation videos                                                                                                                            | 5 weeks                                                |
| Lim [40]                    | Integrative behavioral couple therapy (IBCT): promote acceptance of differences in their relationships and feedback from a psychologist on their completed assignments.                                                                                                                                                                                                                                                                                                                                                                                                           | Web-based program with practice assignments                                                                                                        | 4 days (4 daily sessions, each lasted 20–40 min)       |
| Mirshafieian et al. [41]    | Eight 90-minute solution-focused couples therapy, led by a master’s student in counseling.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Online counseling sessions                                                                                                                         | 8 weeks                                                |
| Nowlan et al. [26]          | Culturally adapted version of OurRelationship program for same-sex couples, same as Group 1 in Hatch et al. [29], which consisted of 3 sections of “Observe,” “Understand,” and “Respond,” with additional personalized feedback on minority stress, relationship concealment, and internalized stigma for those who had LGBTQ-specific concerns.                                                                                                                                                                                                                                 | Online program of 8 h and 15–20 min videoconference calls with a staff coach at the end of each section.                                           | 8 weeks                                                |

**Table 1** (continued)

| Reference            | Components                                                                                                                                                                                                                                                                                                                                           | Delivery formats                                                                                     | Duration |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------|
| Roddy et al. [31]    | OurRelationship program, same as Group 1 in Hatch et al. [29].                                                                                                                                                                                                                                                                                       | Online program of 8 h and interspersed with four 15-minute videoconference calls with a staff coach. | 8 weeks  |
| Rullo et al. [42]    | Sex SMART: Module 1: stress management and resiliency training and read corresponding book chapters from <i>The Mayo Clinic Guide to Stress-Free Living</i> . Module 2: 7 sessions consisting of brief videos and worksheet exercises, where couples learned sexual health psychoeducation, mindfulness and sexual function, sensate-focus training. | Online videos, book chapters, quiz                                                                   | 12 weeks |
| Wittmann et al. [43] | 6 tailored modules addressing expectations for sexual and emotional sequelae of treatment, rehabilitation, and guidance toward sexual intimacy recovery.                                                                                                                                                                                             | Online videos                                                                                        | 28 weeks |

yielding small effects ( $g < 0.2$ ), 39.1% yielding moderate effects ( $0.2 \leq g < 0.8$ ), and 8.7% had large effects ( $g \geq 0.8$ ). Regarding other sex-related outcomes, such as sexual distress, sexual communication, and sexual knowledge, the median effect was 0.66, ranging from 0.15 to 1.76, with 9.1% had small effects ( $g < 0.2$ ), 55% had moderate effects ( $0.2 \leq g < 0.8$ ) and 36.4% had large effects ( $g \geq 0.8$ ). For outcomes about relationship, including relationship satisfaction, communication, emotional intimacy, and relationship problems, the median effect was 0.43, ranging from  $-0.01$  to 1.07, with 28% of the measures yielding small effects ( $g < 0.2$ ), 66% having moderate effects ( $0.2 \leq g < 0.8$ ), and 6% having large effects ( $g \geq 0.8$ ). Importantly, the study by Durbin and colleagues [35] did not explicitly measure relationship or sexuality related outcomes, but investigated the role of physical affection and the enjoyment of physical touch. We included this study in the current review as physical affection and touch were proxy measurements of intimacy.

**Study Population** More than half of the studies focused on couples who had been experiencing some degree of adversity that negatively affected their relationships. Specifically, three studies [27, 36, 42] recruited couples where the female partner was diagnosed with female sexual interest/arousal disorder, two studies targeted on couples with relationship distress [26, 31], and one study focused on couples in “no-sex” or “low-sex” relationships [28]. Among them, in the study by Farahi and colleagues [36], although the effects regarding sexual interest and satisfaction were small ( $g = 0.15$ ,  $0.14$ , respectively), the effects on sexual communication and sexual intercourse frequency were moderate ( $g = 0.5$ ,  $0.66$ , respectively).

On the other hand, some studies recruited couples where one partner was experiencing medical concerns or the family was undergoing psychosocial stressors, such as men diagnosed with localized prostate cancer [43]; postpartum/breastfeeding women [38]; couples with a diagnosis of unexplained infertility [34]; couples with a child over two

years old with Down syndrome [41]; or couples living in poverty line [29]. The remaining studies recruited from the general population with no specific distress [30, 32, 33, 35, 39, 40].

**Intervention Factors** Lastly, it is important to acknowledge that not all interventions prioritized sexual functioning and/or relationship satisfaction. For instance, the primary outcome in Dreischor and colleagues [34] was pregnancy rate with the intervention concentrated on infertility treatment. Some interventions emphasized psychosexual education, reproductive knowledge, and problem-solving, aiming to address broader relationship issues. Also, as discussed in the previous section, the intervention components and duration varied, indicating different levels of intensity. Nevertheless, the intervention duration and intensity did not seem associated with the unified effect sizes. For example, the five-week intervention implemented in Huang and colleagues [32] and Li and colleagues [33] had moderate effect sizes in improving sexual functioning ( $g$  for FSFI =  $0.3$ ,  $g$  for IIEF =  $0.4$ ) and facilitating better sexual communication ( $g$  ranged from  $0.79$  to  $1.61$ ), whereas a four-week weekly group online sexual counseling intervention in Farahi and colleagues [36] did not improve female sexual function ( $g$  for FSFI =  $0.02$ ). Likewise, interventions with longer duration were not necessarily more effective: the intervention conducted in Dreischor and colleagues [34] and Wittmann and colleagues [43] lasted 24 weeks and 28 weeks, respectively, but the outcomes did not yield the greatest effect sizes ( $g$  ranged from  $0.03$  to  $0.29$  for sexual functioning outcomes, and  $0.05$  to  $0.1$  for relationship related outcomes).

In addition, couples' engagement and treatment adherence differed as well, which played a role in the effectiveness of the intervention. As discussed in the previous section, although Roddy and colleagues [31], Mirshafieian and colleagues [41], and Le and colleagues [30] worked with couples experiencing difficult life events and had greater effect sizes for relationship related outcomes, it is crucial to acknowledge that they all incorporated human interactions

**Table 2** Outcomes of reviewed studies

| Reference             | Concept                                          | Name of the measure | Time contrast                                                                  | Ms (SDs)                                                                                                                   | Statistics                                                                | Unified effect size                                                                                                          |
|-----------------------|--------------------------------------------------|---------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Bouchard et al. [27]  | Partner-focused sexual desire in women with SIAD | DSD-P <sup>a</sup>  | Baseline vs. Post-test                                                         | 11.6 (7.1) → 16.9 (7.9)                                                                                                    | $b$ (SE)=5.20 (1.67), $p$ (Time) = .002**, Cohen's $d$ =0.73              | Hedges' $g$ =0.70, 95% CI = [0.20, 1.20], SE=0.26, Variance=0.065 (within-group)                                             |
|                       |                                                  |                     | Baseline vs. Six-month follow-up                                               | 11.6 (7.1) → 17.8 (8.0)                                                                                                    | $b$ (SE)=6.22 (1.71), $p$ (Time) < 0.001***, Cohen's $d$ =0.88            | Hedges' $g$ =0.84, 95% CI = [0.32, 1.36], SE=0.27, Variance=0.071                                                            |
|                       | Sexual distress                                  | SDS-R <sup>b</sup>  | Women with SIAD: Baseline vs. Post-test                                        | 29.5 (9.0) → 20.4 (9.0)                                                                                                    | $b$ (SE) = -8.40 (2.43), $p$ (Time) = 0.001***, Cohen's $d$ = -0.93       | Hedges' $g$ = -0.89, 95% CI = [-1.42, -0.36], SE=0.27, Variance=0.073                                                        |
|                       |                                                  |                     | Baseline vs. Follow-up                                                         | 29.5 (9.0) → 19.3 (8.2)                                                                                                    | $b$ (SE) = -9.88 (2.31), $p$ (Time) < 0.001***, Cohen's $d$ = -1.10       | Hedges' $g$ = -1.05, 95% CI = [-1.61, -0.49], SE=0.29, Variance=0.082                                                        |
|                       |                                                  |                     | Partners: Baseline vs. Post-test                                               | 21.7 (8.8) → 16.9 (10.8)                                                                                                   | $b$ (SE) = -4.79 (1.52), $p$ (Time) = 0.002**, Cohen's $d$ = -0.54        | Hedges' $g$ = -0.52, 95% CI = [-0.99, -0.04], SE=0.24, Variance=0.06                                                         |
|                       |                                                  |                     | Baseline vs. Follow-up                                                         | 21.7 (8.8) → 19.1 (10.8)                                                                                                   | $b$ (SE) = -2.73 (1.33), $p$ (Time) = 0.04*, Cohen's $d$ = -0.31          | Hedges' $g$ = -0.30, 95% CI = [-0.76, 0.17], SE=0.24, Variance=0.057                                                         |
| Dreischor et al. [34] | Female sexual function                           | FSFI <sup>c</sup>   | Baseline vs. Three-month after randomization vs. Six-month after randomization | Intervention group (Int): 28.8 (4.4) → 29.6 (4.0) → 29.2 (4.5); Control group (Ctrl): 29.0 (4.2) → 28.8 (4.9) → 29.3 (4.6) | $p$ (Time) = 0.32, $p$ (Group) = 0.24, $p$ (Time * Intervention) = 0.24   | Hedges' $g$ =0.23, 95% CI = [0.08, 0.38], SE=0.076, Variance=0.0058 (Time*Group interaction effects at baseline-three-month) |
|                       | Erectile function                                | IIEF <sup>d</sup>   | Baseline vs. Three-month vs. Six-month                                         | Int: 65.9 (6.2) → 65.8 (7.3) → 65.8 (6.8); Ctrl: 66.3 (6.1) → 66.0 (6.0) → 64.4 (9.0)                                      | $p$ (Time) < 0.01**, $p$ (Group) = 0.49, $p$ (Time * Intervention) = 0.49 | Hedges' $g$ =0.03, 95% CI = [-0.12, 0.18], SE=0.076, Variance=0.0058                                                         |
|                       | Sexual pleasure (female)                         | QSE <sup>e</sup>    | Baseline vs. Three-month vs. Six-month                                         | Int: 5.4 (0.7) → 5.4 (0.8) → 5.6 (0.8); Ctrl: 5.5 (0.9) → 5.4 (0.9) → 5.6 (1.0)                                            | $p$ (Time) = 0.8, $p$ (Group) = 0.04*, $p$ (Time * Intervention) = 0.04*  | Hedges' $g$ =0.12, 95% CI = [-0.03, 0.27], SE=0.076, Variance=0.0058                                                         |
|                       | Sexual pleasure (male)                           | QSE                 | Baseline vs. Three-month vs. Six-month                                         | Int: 5.6 (0.8) → 5.5 (0.7) → 5.7 (0.8); Ctrl: 5.5 (0.8) → 5.6 (0.7) → 5.6 (1.0)                                            | $p$ (Time) < 0.01**, $p$ (Group) = 0.59, $p$ (Time * Intervention) = 0.59 | Hedges' $g$ = -0.25, 95% CI = [-0.40, -0.10], SE=0.076, Variance=0.0058                                                      |

**Table 2** (continued)

| Reference          | Concept                           | Name of the measure | Time contrast                          | Ms (SDs)                                                                              | Statistics                                                                                                                | Unified effect size                                                                                                                                                                                                                                                                                   |
|--------------------|-----------------------------------|---------------------|----------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | Relational well-being (female)    | R-DAS <sup>f</sup>  | Baseline vs. Three-month vs. Six-month | Int: 52.2 (5.8) → 52.5 (5.2) → 52.6 (5.3); Ctrl: 52.9 (5.5) → 52.7 (6.0) → 52.8 (6.1) | p (Time) = 0.79, p (Group) = 0.16, p (Time * Intervention) = 0.16                                                         | Hedges' g=0.09, 95% CI = [-0.06, 0.24], SE=0.076, Variance=0.0058                                                                                                                                                                                                                                     |
|                    | Relational well-being (male)      | R-DAS               | Baseline vs. Three-month vs. Six-month | Int: 51.2 (6.0) → 51.7 (6.1) → 52.2 (6.4); Ctrl: 51.6 (5.7) → 51.8 (5.9) → 51.1 (6.0) | p (Time) = 0.24, p (Group) = 0.76, p (Time * Intervention) = 0.76                                                         | Hedges' g=0.05, 95% CI = [-0.10, 0.20], SE=0.076, Variance=0.0058                                                                                                                                                                                                                                     |
| Durbin et al. [35] | Momentary intimacy                | N/A                 | N/A                                    | N/A                                                                                   | Group assignment positively impacted momentary intimacy (b = 0.38, <i>p</i> = .04*)                                       | Hedges' g=0.65, 95% CI = [-0.00, 1.29], SE=0.33, Variance=0.108 (a lagged-DV APIM effect, current intimacy ~ group + prior intimacy (+ partner terms), with dyad clustering, so it's a conditional, longitudinal between-group contrast (causal given randomization), adjusted for the prior outcome) |
|                    | Momentary security                | N/A                 | N/A                                    | N/A                                                                                   | Group assignment had no significant impact on momentary security (b = 0.14, <i>p</i> = .430).                             | Hedges' g=0.25, 95% CI = [-0.39, 0.88], SE=0.32, Variance=0.103 (adjusted and conditional on prior values)                                                                                                                                                                                            |
| Farahi et al. [36] | Female sexual interest and desire | SIDI-F <sup>g</sup> | Baseline vs. Post-test                 | Group A: 24.8 (7) → 36.7 (9); Group B: 23.6 (9) → 38.4 (12)                           | Group A: <i>p</i> (Time) < 0.001***, Cohen's d=1.4; Group B: <i>p</i> (Time) < 0.001***, Cohen's d=1.4. p (Group) = 0.520 | Comparing two intervention groups, just that Group B had partner participate with them: Between-group post-test comparison: Hedges' g=0.15, 95% CI = [-0.32, 0.63], SE=0.24, Variance=0.058                                                                                                           |
|                    | Sexual Satisfaction               | GMSS <sup>h</sup>   | Baseline vs. Post-test                 | Group A: 18.8 (5) → 22.5 (4); Group B: 18 (6) → 23.4 (6)                              | Group A: <i>p</i> (Time) < 0.001***, Cohen's d=0.7; Group B: <i>p</i> (Time) < 0.001***, Cohen's d=0.8. p (Group) = 0.547 | Hedges' g=0.14, 95% CI = [-0.33, 0.62], SE=0.24, Variance=0.058                                                                                                                                                                                                                                       |
|                    | Female sexual function            | FSFI-6 <sup>i</sup> | Baseline vs. Post-test                 | Group A: 14 (4) → 19.5 (4); Group B: 13.1 (3) → 19.5 (5)                              | Group A: <i>p</i> (Time) < 0.001***, Cohen's d=1.1; Group B: <i>p</i> (Time) < 0.001***, Cohen's d=1.4. p (Group) = 0.933 | Hedges' g=0.02, 95% CI = [-0.45, 0.49], SE=0.24, Variance=0.058                                                                                                                                                                                                                                       |

**Table 2** (continued)

| Reference           | Concept                         | Name of the measure | Time contrast                                                                                                    | Ms (SDs)                                                                                                                                                                                                                                         | Statistics                                                                                                                                                                                           | Unified effect size                                                                                                                                            |
|---------------------|---------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | Female sexual distress          | FSD-3 <sup>j</sup>  | Baseline vs. Post-test                                                                                           | Group A: 6.8 (2) → 4.2 (2);<br>Group B: 7.4 (2) → 4.4 (2)                                                                                                                                                                                        | Group A: <b><i>p</i> (Time) &lt; 0.001***</b> , Cohen's <i>d</i> =1.2; Group B: <b><i>p</i> (Time) &lt; 0.001***</b> , Cohen's <i>d</i> =1.3. <b><i>p</i> (Group) = 0.542</b>                        | Hedges' <i>g</i> =0.15, 95% CI = [−0.33, 0.62], SE=0.24, Variance=0.058                                                                                        |
|                     | Sexual communication            | DSCS <sup>k</sup>   | Baseline vs. Post-test                                                                                           | Group A: 45 (11) → 50.3 (7);<br>Group B: 45.7 (11) → 54.6 (9)                                                                                                                                                                                    | Group A: <b><i>p</i> (Time) = 0.001***</b> , Cohen's <i>d</i> =0.6; Group B: <b><i>p</i> (Time) &lt; 0.001***</b> , Cohen's <i>d</i> =0.9. <b><i>p</i> (Group) = 0.037*</b> , Cohen's <i>d</i> =0.5. | Hedges' <i>g</i> =0.50, 95% CI = [−0.02, 0.98], SE=0.25, Variance=0.060                                                                                        |
|                     | Sexual dysfunction beliefs      | SDBQ <sup>l</sup>   | Baseline vs. Post-test                                                                                           | Group A: 93 (16) → 90.7 (14);<br>Group B: 91.9 (17) → 85.9 (18)                                                                                                                                                                                  | Group A: <b><i>p</i> (Time) = 0.433</b> ; Group B: <b><i>p</i> (Time) = 0.076. <i>p</i> (Group) = 0.241</b>                                                                                          | Hedges' <i>g</i> =0.28, 95% CI = [−0.20, 0.75], SE=0.24, Variance=0.059                                                                                        |
|                     | Frequency of sexual intercourse | N/A                 | Baseline vs. Post-test                                                                                           | Group A: 4.5 (3) → 6 (2);<br>Group B: 5.3 (5) → 8.3 (3)                                                                                                                                                                                          | Group A: <b><i>p</i> (Time) = 0.005**</b> , Cohen's <i>d</i> =0.5; Group B: <b><i>p</i> (Time) = 0.004**</b> , Cohen's <i>d</i> =0.6. <b><i>p</i> (Group) = 0.006**</b> , Cohen's <i>d</i> =0.7.     | Hedges' <i>g</i> =0.66, 95% CI = [0.17, 1.14], SE=0.25, Variance=0.061                                                                                         |
| Hatch et al. [29]   | Sexual intimacy                 | PAIR <sup>m</sup>   | Baseline vs. Post-test                                                                                           | Sample 1:<br>Group 1: 21.0 (5.5) → 22.7 (5.8);<br>Group 2: 20.6 (5.2) → 22.3 (5.4);<br>Ctrl: 20.3 (5.7) → 19.6 (6.1)<br><br>Sample 2:<br>Group 1: 21.2 (5.5) → 22.5 (5.2);<br>Group 2: 20.8 (5.1) → 22.0 (5.7);<br>Ctrl: 20.8 (5.3) → 20.5 (5.9) | Group 1 vs. Ctrl: <b>95% CI [0.18, 0.41]</b> , Cohen's <i>d</i> =0.28.<br>Group 2 vs. Ctrl: <b>95% CI [0.23, 0.45]</b> , Cohen's <i>d</i> =0.34.                                                     | Hedges' <i>g</i> =0.50, 95% CI = [0.39, 0.61], SE=0.056, Variance=0.003 between-group difference at post-test (adjusted for pre-test via the multilevel slope) |
| Heidari et al. [38] | Female sexual function          | FSFI                | Trajectory of change over baseline, immediately after intervention, one-month follow up, and two-month follow-up | Int: 20.6 (6.1) → 24.4 (1.7) → 24.8 (3.1);<br>Ctrl: 20.8 (3.8) → 21.4 (1.7) → 21.9 (2) → 22 (3.1)                                                                                                                                                | Group effect: <b>F=58.08, <i>p</i> &lt; .001</b> , partial $\eta^2 = 0.39$ (ANCOVA-adjusted for baseline).                                                                                           | Hedges' <i>g</i> =1.75, 95% CI [1.36, 2.14], SE=0.20, Variance=0.039 between-group difference immediately post-intervention (ANCOVA-adjusted for baseline)     |

**Table 2** (continued)

| Reference         | Concept                   | Name of the measure | Time contrast                                        | Ms (SDs)                                                                                                           | Statistics                                                                                       | Unified effect size                                                                                                                                            |
|-------------------|---------------------------|---------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Huang et al. [32] | Erectile function         | IIEF-5              | Trajectory of change over four time points as above. | Int: 20.6 (3.6) → 22.4 (1.3) → 22.7 (1.2) → 22.4 (2.2);<br>Ctrl: 21.2 (3.3) → 21.2 (1.3) → 21.1 (1.2) → 21.1 (2.2) | Group effect: $F = 39.24$ , $p < .001$ , partial $\eta^2 = 0.23$ (ANCOVA-adjusted for baseline). | Hedges' $g = 0.94$ , 95% CI [0.59, 1.29], SE = 0.18, Variance = 0.031 between-group difference immediately post-intervention (ANCOVA-adjusted for baseline)    |
|                   | Female sexual distress    | SDS-R               | Trajectory of change over four time points as above. | Int: 7.5 (7.3) → 4.4 (2.3) → 4.2 (2.0) → 3.1 (2.1);<br>Ctrl: 7.1 (5.1) → 6.8 (2.3) → 6.4 (2.0) → 5.9 (2.1)         | Group effect: $F = 65.75$ , $p < .001$ , partial $\eta^2 = 0.33$ (ANCOVA-adjusted for baseline). | Hedges' $g = -1.00$ , 95% CI [-1.35, -0.65], SE = 0.18, Variance = 0.032 between-group difference immediately post-intervention (ANCOVA-adjusted for baseline) |
|                   | Male sexual distress      | SDS-R               | Trajectory of change over four time points as above. | Int: 8.0 (7.9) → 4.5 (2.5) → 4.1 (3.1) → 3.5 (2.6);<br>Ctrl: 7.8 (7.1) → 7.5 (2.5) → 7.4 (3.1) → 7.1 (2.6)         | Group effect: $F = 56.36$ , $p < .001$ , partial $\eta^2 = 0.30$ (ANCOVA-adjusted for baseline). | Hedges' $g = -1.19$ , 95% CI [-1.55, -0.83], SE = 0.18, Variance = 0.033 between-group difference immediately post-intervention (ANCOVA-adjusted for baseline) |
|                   | Female sexual function    | FSFI                | Baseline vs. Post-test                               | Int: 26.5 (7.0) → 28.3 (5.4);<br>Ctrl: 27.3 (5.1) → 27.2 (5.7)                                                     | $p$ (Time * Intervention) > 0.05                                                                 | Hedges' $g = 0.30$ , 95% CI [-0.32, 0.92], SE = 0.32, Variance = 0.10                                                                                          |
|                   | Erectile function         | IIEF                | Baseline vs. Post-test                               | Int: 21.4 (2.4) → 22.4 (2.1);<br>Ctrl: 20.1 (5.4) → 19.4 (5.7)                                                     | $p$ (Time * Intervention) > 0.05                                                                 | Hedges' $g = 0.40$ , 95% CI [-0.22, 1.02], SE = 0.32, Variance = 0.10                                                                                          |
|                   | Early ejaculation         | CHEES <sup>n</sup>  | Baseline vs. Post-test                               | Int: 10.9 (2.9) → 10.5 (2.6);<br>Ctrl: 9.6 (3.4) → 9 (2.6)                                                         | $p$ (Time * Intervention) > 0.05                                                                 | Hedges' $g = 0.06$ , 95% CI [-0.56, 0.68], SE = 0.32, Variance = 0.10                                                                                          |
|                   | Relationship satisfaction | RAS <sup>o</sup>    | Baseline vs. Post-test                               | Int: 29.5 (4.8) → 29.8 (4.1);<br>Ctrl: 29.7 (4.6) → 29.5 (3.6)                                                     | $p$ (Time * Intervention) > 0.05                                                                 | Hedges' $g = 0.11$ , 95% CI [-0.32, 0.53], SE = 0.22, Variance = 0.048                                                                                         |
|                   | Intimacy                  | PAIR                | Baseline vs. Post-test                               | Int: 141.3 (22.6) → 145.8 (19.9);<br>Ctrl: 152.3 (11.9) → 147.6 (17.6)                                             | $p$ (Time * Intervention) > 0.05                                                                 | Hedges' $g = 0.50$ , 95% CI [0.07, 0.93], SE = 0.22, Variance = 0.048                                                                                          |
|                   | Sexual satisfaction       | GMSS                | Baseline vs. Post-test                               | Int: 58.5 (7) → 58.5 (6.2);<br>Ctrl: 59.1 (6.4) → 58.1 (7.9)                                                       | $p$ (Time * Intervention) > 0.05                                                                 | Hedges' $g = 0.15$ , 95% CI [-0.33, 0.63], SE = 0.25, Variance = 0.060                                                                                         |

**Table 2** (continued)

| Reference                    | Concept                                  | Name of the measure | Time contrast                                                           | Ms (SDs)                                                                                                                                                                                                                                            | Statistics                                                                                                                                                                                                                                                       | Unified effect size                                                                                                  |
|------------------------------|------------------------------------------|---------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| İnan Kırmızıgül & Şahin [39] | Sexual and reproductive health knowledge | SRHP                | Baseline vs. Post-test                                                  | Int: 15.6 (3.4) → 18.5 (3.1);<br>Ctrl: 15.4 (4.4) → 15.2 (4.1)                                                                                                                                                                                      | F=93.65, <b>p (Time * Intervention) &lt; 0.001***</b>                                                                                                                                                                                                            | Hedges' g=1.76, 95% CI = [1.34, 2.18], partial $\eta^2$ = 0.44<br>SE=0.22<br>Variance=0.046                          |
|                              | Marital adjustment                       | MAS <sup>q</sup>    | Post-test                                                               | Int-Female: 50.7 (6.2),<br>Int-Male: 50.2 (6.5);<br>Ctrl-Female: 45.5 (6.7),<br>Ctrl-Male: 46.8 (7.1)                                                                                                                                               | F = 12.63, <b>p (Group) = 0.001***</b>                                                                                                                                                                                                                           | Hedges' g=0.64, 95% CI = [0.28, 1.01], SE=0.19, Variance=0.035 (between-group post-test comparison)                  |
|                              | Sexual satisfaction                      | NSSS <sup>r</sup>   | Post-test                                                               | Int-Female: 80.7 (14.5),<br>Int-Male: 82.4 (14.9);<br>Ctrl-Female: 74.1 (18.5),<br>Ctrl-Male: 74.2 (19.7)                                                                                                                                           | F = 5.61, <b>p (Group) = 0.019*</b>                                                                                                                                                                                                                              | Hedges' g=0.43, 95% CI = [0.07, 0.79], SE=0.18, Variance=0.033                                                       |
|                              | Sexual self-efficacy                     | SSES <sup>s</sup>   | Post-test                                                               | Int-Female: 34.1 (6.4),<br>Int-Male: 35.4 (5.1);<br>Ctrl-Female: 31.5 (6.9),<br>Ctrl-Male: 30.2 (7.5)                                                                                                                                               | F = 10.56, <b>p (Group) = 0.002**</b>                                                                                                                                                                                                                            | Hedges' g=0.59, 95% CI = [0.22, 0.95], SE=0.19, Variance=0.035                                                       |
| Kleinplatz et al. [28]       | Sexual satisfaction                      | NSSS                | Baseline vs. Post-test                                                  | In-person group M change=13.2;<br>Online group M change=10.8                                                                                                                                                                                        | In-person group: t(74)=10.2, <b>p (Time) &lt; 0.001***</b> , Cohen's d=1.2;<br>Online group: t(65)=7.8, <b>p (Time) &lt; 0.001***</b> , Cohen's d=1.0                                                                                                            | Hedges' g=0.21, 95% CI = [-0.12, 0.54], SE=0.17, Variance=0.028                                                      |
| Le et al. [30]               | Relationship satisfaction                | CSI-4 <sup>t</sup>  | Baseline vs. Post-test vs. Two-month follow-up vs. Four-month follow up | Int: 10.1 (4.5) → 13.3 (4.7) → 13.6 (4.6) → 13.7 (4.9) for male,<br>8.2 (4.5) → 12.5 (5.2) → 12.8 (4.8) → 12.9 (5.0) for female;<br>Ctrl: 10.1 (4.7) → 10.9 (4.9) → 10.5 (4.6) → 11.1 (5.3) for male, 8.5 (4.8) → 10 (4.8) → 9.9 (4.7) → 10.4 (4.8) | Intervention status significantly predicts couples' trajectory profile, $\chi^2$ (3)=52.19, p < .001, indicating program effects.<br>OR: The odds of following large improvement trajectory are 7.03 times higher for intervention couples than control couples. | Hedges' g=1.07, 95% CI = [0.90, 1.24], SE=0.087, Variance=0.0076 (between-group difference in trajectory membership) |

**Table 2** (continued)

| Reference      | Concept                                                       | Name of the measure | Time contrast                                  | Ms (SDs)                                                                                                                         | Statistics                                                                          | Unified effect size                                                     |
|----------------|---------------------------------------------------------------|---------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Li et al. [33] | Sexual communication- Individual sexual self-disclosure (SSD) | N/A                 | Baseline vs. Post-test                         | Not available                                                                                                                    | b = -1.45, OR = 0.23, 95% CI [0.05, 1.18], <b>p (Time*Intervention) = 0.039*</b>    | Hedges' g = -0.79, 95% CI = [-1.64, 0.09], SE = 0.23, Variance = 0.052  |
|                | Sexual communication- Reported partner's SSD                  | N/A                 | Baseline vs. Post-test                         | Not available                                                                                                                    | b = -2.07, OR = 0.13, 95% CI [0.03, 0.56], <b>p (Time*Intervention) = 0.003**</b>   | Hedges' g = -1.13, 95% CI = [-1.92, -0.32], SE = 0.24, Variance = 0.056 |
|                | Sexual communication- Post-sex sexual talk                    | N/A                 | Baseline vs. Post-test                         | Not available                                                                                                                    | b = 2.95, OR = 19.15, 95% CI [2.33, 157.14], <b>p (Time*Intervention) = 0.003**</b> | Hedges' g = 1.61, 95% CI = [0.46, 2.76], SE = 0.25, Variance = 0.064    |
| Lim [40]       | Perceived differences                                         | PDDQ <sup>u</sup>   | Baseline vs. Post-test vs. Two-month Follow-up | Clinician-guided group: 50.4 (10.8) → 45.3 (9.3) → 43.2 (9.9); Self-guided group: 49.6 (11.2) → 48.4 (10.5) → 47.2 (9.4)         | F = 3.61, <b>p (Time * Intervention) &lt; 0.05*</b> , partial $\eta^2 = 0.06$       | Hedges' g = -0.35, 95% CI = [-0.88, 0.18], SE = 0.27, Variance = 0.073  |
|                | Distress from differences                                     | PDDQ                | Baseline vs. Post-test vs. Two-month Follow-up | Clinician-guided group: 40.6 (8.8) → 35.8 (7.9) → 33.3 (8.1); Self-guided group: 39.6 (9.3) → 34.6 (9.6) → 35.1 (7.9)            | F = 1.66, <b>p (Time * Intervention) &gt; 0.05</b> , partial $\eta^2 = 0.03$        | Hedges' g = 0.02, 95% CI = [-0.50, 0.54], SE = 0.27, Variance = 0.072   |
|                | Acceptance of differences                                     | ADCRS <sup>v</sup>  | Baseline vs. Post-test vs. Two-month Follow-up | Clinician-guided group: 103.8 (19.5) → 121.0 (14.3) → 117.8 (14.4); Self-guided group: 99.4 (14.1) → 111.8 (15.6) → 109.1 (15.8) | F = 2.17, <b>p (Time * Intervention) &gt; 0.05</b> , partial $\eta^2 = 0.04$        | Hedges' g = 0.28, 95% CI = [-0.25, 0.81], SE = 0.27, Variance = 0.073   |
|                | Quality of couples' relationships-Intimacy                    | CRSS <sup>w</sup>   | Baseline vs. Post-test vs. Two-month Follow-up | Clinician-guided group: 53.7 (9.1) → 56.8 (6.9) → 58.2 (6.3); Self-guided group: 54.4 (10.2) → 56.9 (9.9) → 56.0 (8.2)           | F = 1.20, <b>p (Time * Intervention) &gt; 0.05</b> , partial $\eta^2 = 0.02$        | Hedges' g = 0.06, 95% CI = [-0.46, 0.58], SE = 0.27, Variance = 0.072   |

**Table 2** (continued)

| Reference                | Concept                                                   | Name of the measure | Time contrast                                  | Ms (SDs)                                                                                                                | Statistics                                                      | Unified effect size                                                                                 |
|--------------------------|-----------------------------------------------------------|---------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
|                          | Quality of couples' relationships-Stability               | CRSS                | Baseline vs. Post-test vs. Two-month Follow-up | Clinician-guided group: 47.2 (7.5) → 51.4 (4.7) → 50.7 (5.6); Self-guided group: 44.4 (7.9) → 45.9 (8.3) → 46.5 (7.5)   | F=2.27, p (Time * Intervention) > 0.05, partial $\eta^2$ = 0.04 | Hedges' g=0.35, 95% CI = [-0.19, 0.88], SE=0.27, Variance=0.073                                     |
|                          | Quality of couples' relationships-Satisfaction            | CRSS                | Baseline vs. Post-test vs. Two-month Follow-up | Clinician-guided group: 38.2 (5.5) → 41.0 (2.6) → 40.1 (3.3); Self-guided group: 35.9 (6.0) → 38.1 (5.6) → 37.7 (5.1)   | F=0.33, p (Time * Intervention) > 0.05, partial $\eta^2$ = 0.01 | Hedges' g=0.10, 95% CI = [-0.42, 0.63], SE=0.27, Variance=0.072                                     |
|                          | Patterns of couple interaction-Constructive communication | CPQ-SF <sup>x</sup> | Baseline vs. Post-test vs. Two-month Follow-up | Clinician-guided group: 18.0 (4.8) → 20.1 (3.5) → 20.0 (3.8); Self-guided group: 17.3 (4.1) → 18.6 (4.0) → 17.7 (4.2)   | F=0.99, p (Time * Intervention) > 0.05, partial $\eta^2$ = 0.02 | Hedges' g=0.18, 95% CI = [-0.35, 0.71], SE=0.27, Variance=0.072                                     |
|                          | Patterns of couple interaction-Destructive communication  | CPQ-SF              | Baseline vs. Post-test vs. Two-month Follow-up | Clinician-guided group: 26.8 (9.4) → 20.8 (6.7) → 21.5 (8.0); Self-guided group: 28.4 (8.9) → 26.8 (12.3) → 25.5 (10.5) | F=1.81, p (Time * Intervention) > 0.05, partial $\eta^2$ = 0.03 | Hedges' g = -0.48, 95% CI = [-1.01, 0.06], SE=0.27, Variance=0.074                                  |
|                          | Patterns of couple interaction-Frequency of conflicts     | CFS <sup>y</sup>    | Baseline vs. Post-test vs. Two-month Follow-up | Clinician-guided group: 14.2 (5.3) → 12.0 (4.2) → 12.0 (5.4); Self-guided group: 16.8 (6.4) → 15.7 (7.1) → 15.8 (7.1)   | F=1.95, p (Time * Intervention) > 0.05, partial $\eta^2$ = 0.04 | Hedges' g = -0.18, 95% CI = [-0.71, 0.34], SE=0.27, Variance=0.072                                  |
| Mirshafieian et al. [41] | Marital intimacy                                          | MINQ <sup>z</sup>   | Baseline vs. Post-test vs. Follow-up (women)   | Int: 313.2 (70.9) → 363.3 (47.1) → 370.1 (44.6); Ctrl: 315.9 (41.2) → 334.3 (31.4) → 332.4 (30.4).                      | p (Time) < 0.001, <b>p (Group) = 0.047*</b>                     | Hedges' g=0.73, 95% CI = [0.24, 1.22], SE=0.25, Variance=0.063 (between-group post-test comparison) |

**Table 2** (continued)

| Reference          | Concept                       | Name of the measure                    | Time contrast                                  | Ms (SDs)                                                                                                                                               | Statistics                                                                                                                                                                                                                                                       | Unified effect size                                                                                                          |
|--------------------|-------------------------------|----------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
|                    |                               |                                        | Baseline vs. Post-test vs. Follow-up (partner) | Int: Baseline (62.2) → 359.3 (50.5) → 379 (37.3); Ctrl: 294.5 (35.0) → 314.4 (29) → 319.6 (28.4)                                                       | $p(\text{Time}) > 0.001$ , $p(\text{Group}) = 0.002^{**}$                                                                                                                                                                                                        | The baseline M was reported as 6.283, which does not seem to match the discrepancies. As such, Hedges' g was not calculated. |
| Nowlan et al. [26] | Relationship satisfaction     | CSI-16 <sup>aa</sup>                   | Baseline vs. Post-test                         | Not available                                                                                                                                          | $\gamma(\text{SE}) = 2.25 (0.73)$ , $p(\text{Time}) = 0.004^*$ , Cohen's $d = 0.70$                                                                                                                                                                              | Hedges' $g = 0.68$ , 95% CI = [0.30, 1.07], SE = 0.20, Variance = 0.039 (within-group)                                       |
|                    | Relationship confidence       | 2 items from the Confidence Scale [44] | Baseline vs. Post-test                         | Not available                                                                                                                                          | $\gamma(\text{SE}) = 0.69 (0.35)$ , $p(\text{Time}) = 0.056$ , Cohen's $d = 0.42$                                                                                                                                                                                | Hedges' $g = 0.41$ , 95% CI = [0.07, 0.76], SE = 0.18, Variance = 0.031 (within-group)                                       |
|                    | LGBTQ+ identity Intimacy      | LGBPI <sup>bb</sup>                    | Baseline vs. Post-test                         | Not available                                                                                                                                          | $\gamma(\text{SE}) = 2.76 (0.83)$ , $p(\text{Time}) = 0.003^*$ , Cohen's $d = 0.45$                                                                                                                                                                              | Hedges' $g = 0.44$ , 95% CI = [0.04, 0.84], SE = 0.20, Variance = 0.042 (within-group)                                       |
|                    | Communication during conflict | N/A                                    | Baseline vs. Post-test                         | Not available                                                                                                                                          | $\gamma(\text{SE}) = -5.16 (0.76)$ , $p(\text{Time}) < 0.001^{**}$ , Cohen's $d = -1.07$                                                                                                                                                                         | Hedges' $g = -1.04$ , 95% CI = [-1.51, -0.57], SE = 0.24, Variance = 0.058 (within-group)                                    |
|                    | Emotional intimacy            | N/A                                    | Baseline vs. Post-test                         | Not available                                                                                                                                          | $\gamma(\text{SE}) = 1.48 (0.35)$ , $p(\text{Time}) < 0.001^{**}$ , Cohen's $d = 0.52$                                                                                                                                                                           | Hedges' $g = 0.50$ , 95% CI = [0.07, 0.93], SE = 0.22, Variance = 0.049 (within-group)                                       |
| Roddy et al. [31]  | Relationship satisfaction     | CSI-16                                 | Baseline vs. Post-test                         | Int: 9.1 (3.9) → 12.9 (4.2) for male, 7.6 (3.9) → 11.3 (4.7) for female; Ctrl: 9.4 (4.2) → 10.5 (4.8) for male, 7.4 (4.3) → 8.5 (4.8) for female.      | Couples in the treatment group reported greater increases in relationship satisfaction over time than the control group, $b = 0.304$ , SE = 0.045, $p(\text{Time} * \text{Intervention}) < 0.001^{***}$                                                          | Hedges' $g = 0.65$ , 95% CI = [0.49, 0.81], SE = 0.084, Variance = 0.007                                                     |
|                    | Positive communication        | CPQ_SF                                 | Baseline vs. Post-test                         | Int: 13.6 (4.5) → 15.1 (5.5) for male, 12.2 (4.9) → 14.3 (5.4) for female; Ctrl: 13.7 (5.2) → 14.0 (5.5) for male, 11.9 (5.8) → 12.3 (5.7) for female. | Couples in the intervention group endorsed greater increases in positive communication, and greater decreases in negative communication compared to those in the control condition. $b = 0.123$ , SE = 0.057, $p(\text{Time} * \text{Intervention}) = 0.031^*$ ; | Hedges' $g = 0.28$ , 95% CI = [0.12, 0.44], SE = 0.082, Variance = 0.0067                                                    |

**Table 2** (continued)

| Reference | Concept                           | Name of the measure | Time contrast          | Ms (SDs)                                                                                                                                                     | Statistics                                                                                                                                                                                                                                                                  | Unified effect size                                                             |
|-----------|-----------------------------------|---------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
|           | Negative communication            | CPQ_SF              | Baseline vs. Post-test | Int: 24.1 (9.3) → 18.0 (8.4) for male, 28.3 (10.8) → 20.2 (9.7) for female; Ctrl: 25.9 (11.2) → 22.5 (10.8) for male, 29.3 (11.1) → 26.8 (11.3) for females. | $b = -0.468$ , $SE = 0.098$ , $p$ ( <b>Time * Intervention</b> ) < <b>0.001***</b>                                                                                                                                                                                          | Hedges' $g = -0.39$ , 95% CI = [-0.55, -0.23], $SE = 0.082$ , Variance = 0.0068 |
|           | Emotional intimacy                | PAIR                | Baseline vs. Post-test | Int: 16.3 (4.3) → 19.2 (4.4) for male, 13.7 (4.0) → 18.6 (4.2) for female; Ctrl: 16.7 (4.5) → 17.3 (4.7) for male, 15.2 (4.4) → 16.4 (4.9) for female.       | Couples in the intervention group reported a significantly greater increase in emotional intimacy. $b = 0.238$ , $SE = 0.057$ , $p$ ( <b>Time * Intervention</b> ) < <b>0.001***</b>                                                                                        | Hedges' $g = 0.70$ , 95% CI = [0.53, 0.86], $SE = 0.084$ , Variance = 0.0071    |
|           | Self-protective orientation       | SG <sup>cc</sup>    | Baseline vs. Post-test | Int: 10.5 (4.0) → 8.9 (3.9) for male, 11.1 (4.1) → 8.6 (3.8) for female; Ctrl: 11.2 (4.2) → 10.2 (4.4) for male, 11.7 (4.1) → 10.9 (4.3) for female.         | Couples in the intervention group reported a decrease in self-protective orientation. $b = -0.125$ , $SE = 0.048$ , $p$ ( <b>Time * Intervention</b> ) = <b>0.009**</b>                                                                                                     | Hedges' $g = 0.70$ , 95% CI = [0.53, 0.86], $SE = 0.084$ , Variance = 0.0071    |
|           | Relationship problem - Confidence | N/A                 | Baseline vs. Post-test | Int: 4.2 (1.9) → 5.9 (1.1) for male, 3.4 (1.9) → 5.4 (1.5) for female; Ctrl: 4.1 (2.1) → 4.3 (2.1) for male, 3.4 (2.0) → 3.9 (2.1) for female.               | Couples in the treatment group exhibited greater increases over time in relationship problem confidence; a greater decrease in relationship problem severity and residualized acceptance. $b = 0.221$ , $SE = 0.028$ , $p$ ( <b>Time * Intervention</b> ) < <b>0.001***</b> | Hedges' $g = 0.76$ , 95% CI = [0.59, 0.92], $SE = 0.085$ , Variance = 0.0071    |

**Table 2** (continued)

| Reference         | Concept                           | Name of the measure | Time contrast                    | Ms (SDs)                                                                                                                                           | Statistics                                                           | Unified effect size                                                       |
|-------------------|-----------------------------------|---------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------|
| Rullo et al. [42] | Relationship problem - Severity   | N/A                 | Baseline vs. Post-test           | Int: 5.2 (1.2) → 3.6 (1.5) for male, 5.8 (1.2) → 3.9 (1.5) for female; Ctrl: 5.3 (1.4) → 4.4 (1.5) for male, 6.0 (1.0) → 5.2 (1.3) for female.     | b = -0.070, SE = 0.018, <b>p (Time * Intervention) &lt; 0.001***</b> | Hedges' g = -0.74, 95% CI = [-0.91, -0.58], SE = 0.084, Variance = 0.0071 |
|                   | Relationship problem - Acceptance | N/A                 | Baseline vs. Post-test           | Int: -0.3 (1.9) → -0.7 (2.9) for male, 0.3 (1.9) → -0.3 (2.7) for female; Ctrl: -0.5 (2.2) → 0.4 (2.8) for male, 0.4 (1.7) → 0.6 (2.6) for female. | b = -0.096, SE = 0.036, <b>p (Time * Intervention) = 0.008**</b>     | Hedges' g = -0.54, 95% CI = [-0.71, -0.38], SE = 0.083, Variance = 0.0069 |
|                   | Female sexual function            | FSFI                | Baseline vs. Post-test           | Int: 17.2 (5.4) → 20.2 (4.3); Ctrl: 16.4 (5.4) → 18.4 (5.9)                                                                                        | p (Time*Intervention) = 0.13                                         | Hedges' g = 0.18, 95% CI = [-0.33, 0.69], SE = 0.26, Variance = 0.067     |
|                   | Sexual desire                     | SDI-2 <sup>dd</sup> | Baseline vs. Post-test (female)  | Int: 31.2 (15.3) → 37.8 (13.7); Ctrl: 27.2 (16.0) → 35.2 (18.7)                                                                                    | p (Time*Intervention) = 0.57                                         | Hedges' g = -0.09, 95% CI = [-0.60, 0.42], SE = 0.26, Variance = 0.067    |
|                   |                                   |                     | Baseline vs. Post-test (partner) | Int: 61.7 (14.7) → 62.0 (14.1); Ctrl: 66.1 (15.3) → 66.1 (13.9)                                                                                    | p (Time*Intervention) = 0.69                                         | Hedges' g = 0.02, 95% CI = [-0.50, 0.53], SE = 0.26, Variance = 0.067     |
|                   | Relationship satisfaction         | R-DAS               | Baseline vs. Post-test (female)  | Int: 48.8 (6.7) → 48.0 (6.5); Ctrl: 45.6 (8.0) → 44.9 (8.3)                                                                                        | p (Time*Intervention) = 0.28                                         | Hedges' g = -0.01, 95% CI = [-0.52, 0.49], SE = 0.26, Variance = 0.067    |
|                   |                                   |                     | Baseline vs. Post-test (partner) | Int: 48.7 (5.7) → 48.6 (6.3); Ctrl: 45.3 (6.7) → 45.1 (7.6)                                                                                        | p (Time*Intervention) = 0.33                                         | Hedges' g = 0.02, 95% CI = [-0.49, 0.52], SE = 0.26, Variance = 0.067     |
|                   | Sexual distress                   | SDS-R               | Baseline vs. Post-test (female)  | Int: 27.9 (9.5) → 23.4 (9.9); Ctrl: 22.7 (10.9) → 24.9 (10.9)                                                                                      | p (Time*Intervention) = 0.33                                         | Hedges' g = -0.65, 95% CI = [-1.17, -0.13], SE = 0.27, Variance = 0.07    |

**Table 2** (continued)

| Reference            | Concept                               | Name of the measure       | Time contrast                                                          | Ms (SDs)                                                       | Statistics                                                                                                 | Unified effect size                                                                                  |
|----------------------|---------------------------------------|---------------------------|------------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Wittmann et al. [43] | Erectile function                     | IIEF                      | Baseline vs. Post-test (partner)                                       | Int: 25.8 (8.8) → 28.2 (5.5);<br>Ctrl: 25.8 (8.8) → 25.9 (8.6) | <b>p (Time*Intervention) = 0.04**</b>                                                                      | Hedges' g=0.26, 95% CI = [-0.25, 0.77], SE=0.26, Variance=0.067                                      |
|                      | Sexual satisfaction                   | PRMOIS-GSSL <sup>ee</sup> | Baseline vs. Six-month after treatment (patients with prostate cancer) | Int: 62 → 53;<br>Ctrl: 60 → 51                                 | Model Estimate: Arm Effect at 6 month Controlling for Baseline=1.3, 95% CI [-1.6, 4.2], p (Group) = 0.4    | Hedges' g=0.15, 95% CI = [-0.18, 0.48], SE=0.17, Variance=0.028 (between-group post-test comparison) |
|                      |                                       |                           | Baseline vs. Six-month after treatment (partner)                       | Int: 58 → 53;<br>Ctrl: 60 → 41                                 | Model Estimate: Arm Effect at 6 month Controlling for Baseline = -1.3, 95% CI [-4.7, 2.2], p (Group) = 0.5 | Hedges' g = -0.12, 95% CI = [-0.45, 0.21], SE=0.17, Variance=0.028                                   |
|                      | Interest in sex                       | PROMIS-ISL <sup>ff</sup>  | Baseline vs. Three-month (patients)                                    | Int: 60 → 56;<br>Ctrl: 58 → 53                                 | Arm Effect=2.3, 95% CI [-0.8, 5.4], p (Group) = 0.15                                                       | Hedges' g=0.25, 95% CI = [-0.09, 0.58], SE=0.17, Variance=0.029                                      |
|                      |                                       |                           | Baseline vs. Six-month (patients)                                      | Int: 60 → 56;<br>Ctrl: 58 → 53                                 | Arm Effect=1.9, 95% CI [-1.3, 5.1], p (Group) = 0.2                                                        | Hedges' g=0.20, 95% CI = [-0.14, 0.53], SE=0.17, Variance=0.028                                      |
|                      |                                       |                           | Baseline vs. Three-month (partner)                                     | Int: 53 → 50;<br>Ctrl: 53 → 51                                 | Arm Effect=0.01, 95% CI [-2.8, 2.9], p (Group) = 1.0                                                       | Hedges' g=0.001, 95% CI = [-0.33, 0.33], SE=0.17, Variance=0.028                                     |
|                      |                                       |                           | Baseline vs. Six-month (partner)                                       | Int: 53 → 52;<br>Ctrl: 53 → 51                                 | Arm Effect=0.9, 95% CI [-1.6, 3.4], p (Group) = 0.5                                                        | Hedges' g=0.12, 95% CI = [-0.21, 0.45], SE=0.17, Variance=0.028                                      |
|                      | Patient-Quality of life-Sexual domain | EPIC-26 <sup>gg</sup>     | Baseline vs. Three-month                                               | Int: 80 → 36;<br>Ctrl: 79 → 33                                 | Arm Effect=1.9, 95% CI [-10, 14], p (Group) = 0.8                                                          | Hedges' g=0.05, 95% CI = [-0.28, 0.38], SE=0.17, Variance=0.028                                      |
|                      |                                       |                           | Baseline vs. Six-month                                                 | Int: 80 → 41;<br>Ctrl: 79 → 42                                 | Arm Effect = -4.5, 95% CI [-15, 6.3], p (Group) = 0.4                                                      | Hedges' g = -0.14, 95% CI = [-0.47, 0.19], SE=0.17, Variance=0.028                                   |

**Table 2** (continued)

| Reference | Concept                 | Name of the measure | Time contrast            | Ms (SDs)                    | Statistics                                           | Unified effect size                                             |
|-----------|-------------------------|---------------------|--------------------------|-----------------------------|------------------------------------------------------|-----------------------------------------------------------------|
|           | Partner-Sexual function | FSFI                | Baseline vs. Three-month | Int: 24 → 22; Ctrl: 22 → 18 | Arm Effect=2.5, 95% CI [-0.3, 5.4], p (Group) = 0.09 | Hedges' g=0.29, 95% CI = [-0.04, 0.62], SE=0.17, Variance=0.029 |
|           |                         |                     | Baseline vs. Six-month   | Int: 24 → 21; Ctrl: 22 → 18 | Arm Effect=2.1, 95% CI [-0.8, 5.0], p (Group) = 0.2  | Hedges' g=0.24, 95% CI = [-0.09, 0.57], SE=0.17, Variance=0.028 |

\*  $p < .05$ . \*\*  $p < .01$ . \*\*\*  $p < .001$ . Highlights in bold indicate that the result was significant

Full names of the measures:

- a. DSD-P: The partner subscale of the Sexual Desire Inventory-2 [45, 46]
- b. SDS-R: The Sexual Distress Scale-Revised [47]
- c. FSFI: The Female Sexual Function Index [48, 49]
- d. IIEF: The International Index of Erectile Function [50]
- e. QSE: The Quality of Sexual Experience [51]
- f. R-DAS: The Revised Dyadic Adjustment Scale [52]
- g. SIDI-F: The Sexual Interest and Desire Inventory-Female [53]
- h. GMSS: The Global Measure of Sexual Satisfaction [54]
- i. FSFI-6: The 6-item Female Sexual Function Index [55]
- j. FSD-3: Female Sexual Distress Scale-3 [47, 56].
- k. DSCS: The Dyadic Sexual Communication Scale [57, 58]
- l. SDBQ: The Sexual Dysfunction Beliefs Questionnaire [59, 60]
- m. PAIR: The Personal Assessment of Intimacy in Relationship [61]
- n. CHEES: The Checklist for Early Ejaculation Symptoms [62]
- o. RAS: The Relationship Assessment Scale [63]
- p. SRH: The Sexual and Reproductive Health Knowledge Test for Premarital Couples [64]
- q. MAS: The Marital Adjustment Scale [65]
- r. NSSS: The New Sexual Satisfaction Scale [66]
- s. SSES: The Sexual Self-efficacy Scale [67]
- t. CSI-4: The Couples Satisfaction Index 4-item version [68]
- u. PDDQ: The Perceived Differences and Distress Questionnaire [69]
- v. ADCRS: The Acceptance for Differences in Couple Relationships Scale [69]
- w. CRSS: Quality of couples' relationships-Intimacy [70]
- x. CPQ-SF: The Communication Patterns Questionnaire-Short Form [71]
- y. CFS: The Conflict Frequency Scale [72]
- z. MINQ: The Marital Intimacy Needs Questionnaire [73]
- aa. CSI-16: The Couples Satisfaction Index 16-item version [68]
- bb. LGBPI: The Intimacy Subscale of the Lesbian, Gay, and Bisexual Positive Identity Measure [74]
- cc. SG: Four high-loading items from the measure of self-goals [75]
- dd. SDI-2: The Sexual Desire Inventory-2 [46]
- ee. PRMOIS-GSSL: The Patient Reported Outcomes Measurement Information System (PROMIS) Global Satisfaction With Sex Life (GSSL) scale [76]
- ff. PROMIS-ISL: The PROMIS assessments of Interest in Sex Life [76]
- gg. EPIC-26: The Expanded Prostate Cancer Index Composite [77]

in the intervention, such as coach calls, or online couple therapy sessions led by a therapist. Similarly, the study conducted by Heidari and colleagues [38] showed the largest

effect size for sexual function improvements (measured by FSFI and IIEF), but their intervention was online synchronous counseling sessions led by a counselor.

## Discussion

The present review demonstrates that, overall, the effectiveness of online intervention for improving couples' sexual functioning and relationship satisfaction varied across population, measured outcomes, and the intervention techniques. Specifically, online interventions appeared more effective at improving sexual communication and relationship satisfaction than at directly resolving sexual dysfunction symptoms. However, given that sexual communication is itself a core mechanism through which couples negotiate and manage sexual difficulties, these findings suggest that online interventions may have indirect effects on sexual dysfunction, such as by strengthening the interpersonal processes (e.g., communication, relationship satisfaction, and emotional intimacy) that support sexual function, rather than by directly targeting the symptoms themselves.

First, the varying results may be explained by the variation in study population. Eleven out of the seventeen studies focused on couples experiencing some life adversity or met the diagnostic criteria for a sexual dysfunction, whereas others recruited participants from the general population. This matters because the responsiveness to interventions was likely to be different from those experiencing difficult life situations. Indeed, for those couples experiencing major life challenges, the online interventions appeared to be particularly effective in improving the relationship related outcomes, such as increased relationship satisfaction and intimacy, as well as reduced relationship problem severity, evident by  $g \geq 0.7$  for those measures in Roddy and colleagues [31], Mirshafieian and colleagues [41], and Le and colleagues [30].

Secondly, the intervention factors may have played a role, since the frequency of sessions, duration, and the intervention strategies (i.e., the degree of human/therapist involvement; video vs. audio; synchronous sessions vs. self-help programs) differ by studies. For example, eleven studies had some form of human interaction during an online intervention, ranging from direct therapist-led sessions to phone calls or emails to answer participants' questions, whereas the remainder of the studies utilized automatic, self-help online videos or programs. However, it is important to highlight the nuances of the professionals who led the intervention sessions. Those delivered by trained coaches, such as OurRelationship/ePREP-based programs [26, 29, 31], seemed to have moderate-to-large improvements in relationship-related outcomes (satisfaction and problem-solving,  $g=0.50$ – $1.07$ ) and emotional intimacy ( $g=0.50$ – $0.7$ ). Yet when coach contact was limited or asynchronous, as in Dreischor and colleagues' [34] email-based coaching, effects on sexual function were relevantly small (FSFI  $g=0.23$ ; IIEF  $g=0.03$ ). This suggests that immediacy of coach contact may have played a role for intervention

outcomes. On the other hand, interventions led by licensed therapists or counselors produced some of the largest effect sizes observed in the current review, particularly for sexual functioning and distress outcomes. Heidari and colleagues' [38] counselor-led synchronous sessions yielded the largest effects (FSFI  $g=1.75$ ; IIEF-5  $g=0.94$ ; sexual distress  $g=-1.00$  for female and  $-1.19$  for male), and Bouchard and colleagues' [27] therapist-delivered cognitive-behavioral couple therapy showed large decrease in sexual distress ( $g=-0.89$  for female and  $-0.52$  for their partner), as well as increase in sexual desire ( $g=0.70$ ). Yet, this pattern was not consistent across all reviewed studies. For instance, Farahi and colleagues' [36] counselor-led group sessions yielded only small effects on sexual interest and satisfaction ( $g=0.14$ – $0.15$ ), and so did Kleinplatz and colleagues' [28] therapist-led group therapy for sexual satisfaction ( $g=0.21$ ). Taken together, these findings suggest that professional clinical involvement may offer an advantage specifically for addressing sexual dysfunction and distress, while coach-facilitated models appear effective in strengthening relational and communicative outcomes. Overall, human interactions may have served as an opportunity to address questions of the intervention, or to provide a more tailored approach depending on the participants' needs, likely affecting the intervention outcomes.

Third, the intervention modality also seems to influence which outcomes improved. Cognitive-behavioral based strategies were especially helpful in addressing sexual distress [27], and sensate focus techniques were effective in improving sexual functioning and facilitating more positive sexual communication [32, 33]. Psychoeducation, by contrast, demonstrated varying levels of effects depending on what it was paired with. Related, two studies [40, 43] asked participants to rate their satisfaction with the intervention program, and a small number of participants (0% to 19%) in the study by Wittmann and colleagues [43] found the modules "confusing." Therefore, these variations in intervention content likely influenced how participants engaged with the programs, which may explain the wide discrepancies in the effectiveness of these identified online interventions.

Nevertheless, it is important to acknowledge that the studies in this review covered a wide range of demographics. While North American and European populations were the majority, a large minority (41%) were conducted in Asia. In addition, 35% of the studies included LGBTQ+ couples; although the interventions were not specifically designed for LGBTQ+ couples, they were tailored to individual couples' needs. Specifically, Hatch and colleagues [29] found that same-gender relationship status did not affect the intervention outcomes, suggesting that the online inventions have the potential to be adapted for and benefit couples including individuals with diverse gender identities and sexual orientations. Indeed, Nowlan and colleagues, who exclusively

focused on same-sex couples, found that the adaptation of the heteronormative program was culturally sensitive, promising, and flexible, suggesting that future studies could build on existing interventions designed for mix-gender couples, but include LGBTQ-specific topics (e.g., addressing specific minority stress) [26]. This may reduce barriers for sexual and gender diverse couples in accessing needed care due to geographic constraints, stigma, and a lack of affirming providers.

In conclusion, several factors related to intervention implementation, outcome measurements, and target populations influenced the effectiveness of online interventions in improving couples' sexual functioning and relationship satisfaction. This is consistent with the growing concerns in implementation science concerning effect sizes from small, underpowered RCTs not matching other RCTs [78], and that even when certain interventions are deemed to be effective, they are rarely transformed into scalable programs and reach the general public [79]. Further, some interventions that demonstrate statistical significance in RCTs do not demonstrate clinically significant effects and thus fail to benefit those who are in need of treatment [80]. Therefore, as online interventions have considerable advantages in cost-effectiveness and accessibility, further studies are warranted in investigating why and how interventions work. Based on the NIH Stage Model [79], understanding the mechanisms of change is an integrated aspect that is essential across all stages of intervention development. It is paramount to identify the components that are most effective in improving sexual functioning and relationship satisfaction, as this would inform us how interventions should be generated. This would also inform next stages about how to simplify the intervention to save costs, as well as provide insights into how interventions can be tailored and adapted into different populations for wider implementation.

## Key References

Dreischor, F., Laan, E. T. M., Apers, S., Repping, S., van Lunsen, R. H. W., Lambalk, C. B., D'Hooghe, T. M., Goddijn, M., Custers, I. M., & Dancet, E. A. F. (2020). The stepwise development of an interactive web-based sex education programme for subfertile couples: The Pleasure & Pregnancy programme. *Human Reproduction*, 35(8), 1839–1854. <https://doi.org/10.1093/humrep/deaa106>

- This article describes an innovative intervention that is interactive, and web based, focusing on couple-level mechanisms such as communication,

mindfulness exercises, and sensate focus. It highlights rationales for further research in dyadic, digitally delivered sexual intervention since it is easier to be implemented widely, reducing barriers for those who cannot attend in-person.

Briken, P., Matthiesen, S., Pietras, L., Wiessner, C., Klein, V., Reed, G. M., & Dekker, A. (2020). Estimating the Prevalence of Sexual Dysfunction Using the New ICD-11 Guidelines. *Dtsch Arztebl Int*, 117(39), 653–658. <https://doi.org/10.3238/arztebl.2020.0653>

- This study provides the ICD-11 - based prevalence estimates of sexual dysfunction using a nationally representative sample in Germany, which underscores the importance of sexual dysfunction as an obstacle to sexual health.

Beaulieu, N., Bergeron, S., Brassard, A., Byers, E. S., & Pélouquin, K. (2023). Toward an Integrative Model of Intimacy, Sexual Satisfaction, and Relationship Satisfaction: A Prospective Study in Long-Term Couples. *The Journal of Sex Research*, 60(8), 1100–1112. <https://doi.org/10.1080/00224499.2022.2129557>

- This longitudinal dyadic study shows bidirectional links between partners' intimacy behaviors and both sexual and relationship satisfaction, where intimacy could reinforce satisfaction, and higher satisfaction could also promote intimacy over time. These findings provide a strong basis for future research investigating the intertwining associations between sexual wellbeing and relationship quality in long-term couples.

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## Declarations

**Human and Animal Rights** This article does not contain any studies with human or animal subjects performed by any of the authors.

**Competing interests** The authors declare no competing interests.

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